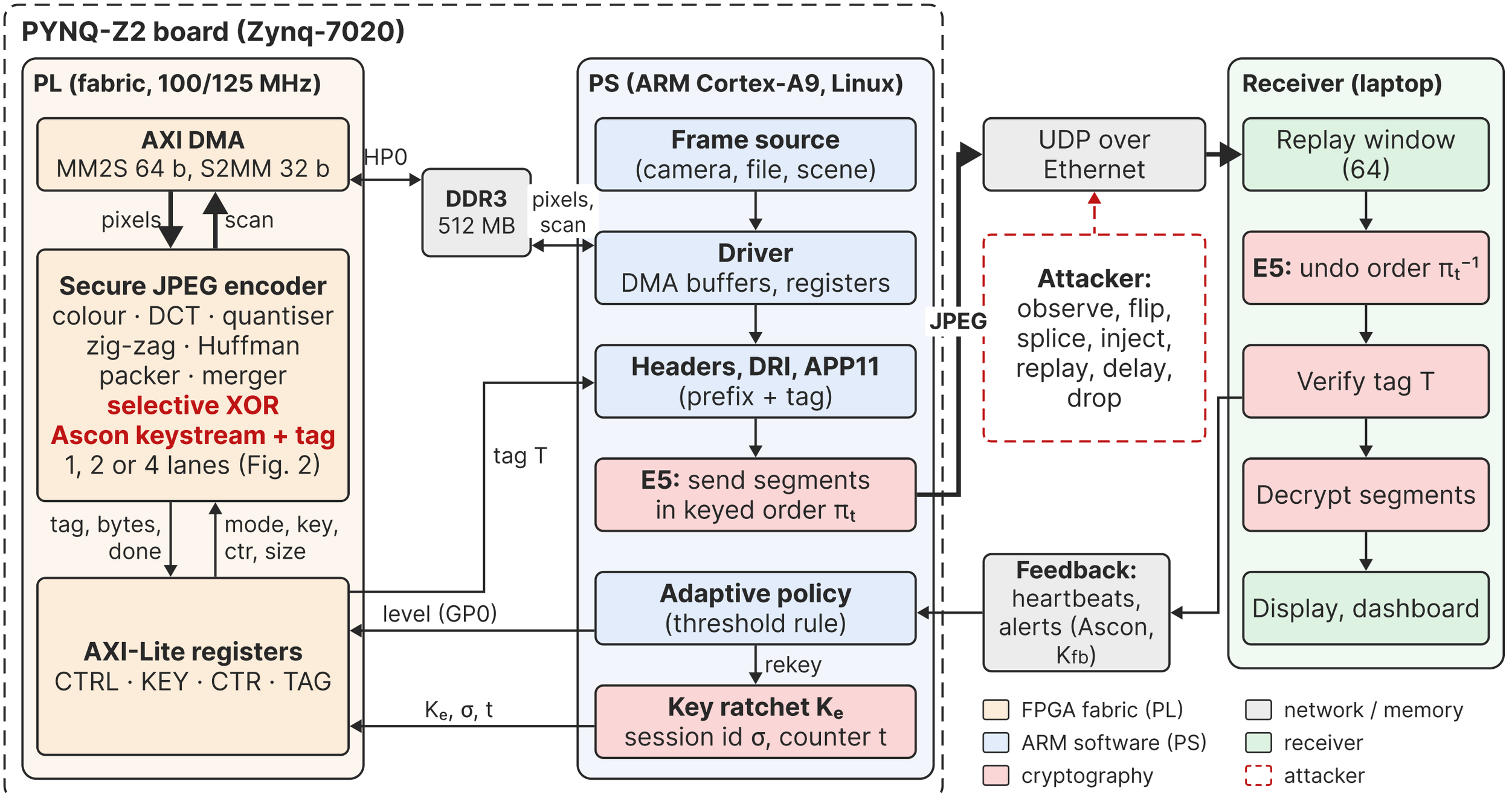

PYNQ-Z2 board (Zynq-7020)
PL (fabric, 100/125 MHz)
AXI DMA
MM2S 64 b, S2MM 32 b
HP0
DDR3
512 MB
pixels
scan
pixels, scan
Secure JPEG encoder
colour · DCT · quantiser
zig-zag · Huffman
packer · merger
selective XOR
Ascon keystream + tag
1, 2 or 4 lanes (Fig. 2)
tag, bytes, done
mode, key, ctr, size
tag T
AXI-Lite registers
CTRL · KEY · CTR · TAG
level (GP0)
Ke, σ, t
PS (ARM Cortex-A9, Linux)
Frame source
(camera, file, scene)
Driver
DMA buffers, registers
Headers, DRI, APP11
(prefix + tag)
E5: send segments
in keyed order πt
Adaptive policy
(threshold rule)
rekey
Key ratchet Ke
session id σ, counter t
JPEG
UDP over Ethernet
Attacker:
observe, flip, splice, inject, replay, delay, drop
Feedback:
heartbeats, alerts (Ascon, Kfb)
Receiver (laptop)
Replay window (64)
E5: undo order πt−1
Verify tag T
Decrypt segments
Display, dashboard
FPGA fabric (PL)
ARM software (PS)
cryptography
network / memory
receiver
attacker